



SNG BUILDER SERIES — GUIDE #003 OF 4

52KW CRF450 ELECTRIC BUILD

FULL BUILD MANUAL — FROM FRAME TO 70HP

52kW

PEAK POWER

70HP

EQUIVALENT
OUTPUT

<\$9K

TOTAL PARTS
COST

450

FRAME
PLATFORM

Start with a CRF450 frame. Follow this guide exactly. End with a 52kW electric machine that costs more new than it cost you to build.

Don't have a 450? Doesn't matter. Sourcing everything on eBay comes out to the same cost either way. This guide works from zero.

WHAT'S INSIDE THIS GUIDE

Platform Overview

CRF450 frame selection, why it works, CRF plastics fitment on extended geometry

Full Parts List

Every component, sourced and priced — motor, controller, battery, suspension, plastics, fasteners

Motor & Controller Setup

52kW motor selection, controller pairing, programming baseline

Battery Pack Configuration

Cell selection, BMS setup, enclosure, thermal management

CRF Plastics Fitment

Extended geometry mounting, trimming notes, hardware, panel fitment sequence

Wiring & Electrical

Full wiring diagram, connector types, signal routing, throttle calibration

Assembly Sequence

Step-by-step build order, torque specs, critical checkpoints

First Start & Tuning

Initial power-on checklist, controller tuning, first ride setup

BUILDER'S NOTE — BRYAN JAWORSKI, SNG CYCLES

This is the build I wished existed when I started. Every part number, every fitment note, every mistake I made so you don't have to — it's all in here. The 450 frame gives you geometry and suspension travel the Sur-Ron platforms can't touch. CRF plastics — it's a CRF frame, use CRF plastics, they fit right and look right. And 52kW for under \$9K is real — I have the receipts.



BUILD SPECIFICATIONS

52kW

PEAK MOTOR POWER

70HP

HP EQUIVALENT

<\$9K

PARTS BUDGET

450

FRAME CLASS

PLATFORM	
Base Frame	CRF520E / CRF450 — sourced on eBay or salvage
Swingarm	3.5" Extended
Body Plastics	CRF — correct fitment for the CRF frame, right mounting points, right lines
Wheels Front + Rear	KKE 17" Supermoto — Gold (new)
Brakes	Front & Rear Hydraulic
Top Speed	85+ MPH
Weight (Est.)	~250 lbs
DRIVETRAIN	
Motor	Ventus 72V 52kW (E-520)
Controller	Ventus 72V
Primary Reduction	12T / 48T — 4:1
Secondary Reduction	14T / 54T — 3.85:1
Total Reduction	15.44:1 (Dual-Stage Jackshaft)
Chain	420 Chain Conversion
ELECTRICAL	
Battery	72V High Performance Lithium
Lighting	Baja Designs — full kit
BUILD ECONOMICS	
Total Build Cost	\$8,659+ USD
Build Time	Experienced builder: 60–80 hrs. First-time: 100–120 hrs with this guide.

WHY THE CRF450 FRAME?

The 450-class MX frame gives you things that other electric build platforms can't: full-travel MX suspension from the factory, a proven geometry designed for aggressive riding, and a parts ecosystem that's everywhere. Surplus frames and donor bikes are plentiful on eBay — and they're cheap because the ICE powerplant is gone or worn out, which doesn't matter to you.

CRF plastics are the right call on this build — you are working with a CRF frame, they fit correctly, the mounting points line up, and the lines look right. The exact fitment and trimming spec is documented step-by-step in the guide.

THE REAL MATH

A 52kW electric motorcycle bought retail costs \$12,000–\$20,000+. This build hits that power level for under \$9K in parts — documented, sourced, and priced in this guide. The knowledge is what you're buying. The parts are on eBay right now.

WHAT YOU NEED BEFORE YOU START

PREREQUISITES	
Frame / Donor Bike	CRF450 frame (bare or complete donor — ICE removed if complete). See guide for year compatibility.
Tools	Basic hand tools, multimeter, torque wrench (metric), angle grinder, drill — full list in guide.
Skill Level	Intermediate. Mechanical comfort required. Electrical wiring comfort required. Welding not needed.
Budget	Under \$9,000 all-in. Parts list is priced as of build date — eBay pricing, not MSRP.
Space	Standard 1-car garage. No lift required — work stands included in tool list.



COMPLETE SOURCED PARTS LIST

SOURCING NOTES

All parts priced as sourced — primarily eBay, with select specialty suppliers noted. Prices reflect current market at time of build. eBay search terms included for each category. Total target: under \$9,000.

COMPONENT	QTY	SPEC / NOTES	SOURCE	EST. COST
FRAME & SUSPENSION				
CRF450 Frame	1	Bare or donor — year compatibility in guide	eBay	See guide
Front Forks	1 set	OEM or upgraded — spec in guide	eBay / OEM	See guide
Swingarm	1	Stock CRF450 retained	Donor	—
Rear Shock	1	OEM or aftermarket — spec in guide	eBay	See guide
Linkage Set	1 set	OEM retained	Donor	—
Steering Stem / Head Bearings	1 set	Replace if donor — spec in guide	eBay	See guide
MOTOR & DRIVETRAIN				
Electric Motor — 52kW	1	Full spec, part number, supplier in guide	Specialty	See guide
Motor Controller	1	Matched to motor — programming baseline included	Specialty	See guide
Sprockets (Primary)	1 set	Per guide ratio spec	eBay	See guide
Sprockets (Secondary)	1 set	Per guide ratio spec	eBay	See guide
Chain — Primary	1	#420 — length in guide	eBay	See guide
Chain — Secondary / Final	1	Per guide spec	eBay	See guide
Jackshaft Assembly	1	See Guide #004 (E-520 Jackshaft) for full fab spec	Fabricated	~\$200-400
BATTERY SYSTEM				
Battery Cells	Per spec	Cell type, voltage, capacity — full spec in guide	Specialty	See guide
BMS (Battery Mgmt System)	1	Matched to cell spec — wiring diagram in guide	Specialty	See guide
Battery Enclosure / Tray	1	Fabricated — dimensions and material in guide	Fabricated	See guide
Battery Connectors	Per spec	Type and gauge in guide	eBay	See guide
On-Board Charger	1	Spec and plug type in guide	Specialty	See guide
BODY & PLASTICS				
CRF Plastic Kit	1 set	Year / model compatibility in guide · correct fit for CRF frame	eBay	See guide
Seat	1	OEM or aftermarket — mounting spec in guide	eBay	See guide
Shrouds / Side Panels	1 set	Trimming notes in guide — where to cut, what to keep	CRF kit	—
ELECTRICAL & CONTROLS				
Throttle Assembly	1	Type and calibration procedure in guide	eBay	See guide
Wiring Harness	1	Custom — full schematic in guide. Wire gauge and connector specs included.	Custom build	See guide
Display / Dash	1	Optional — spec if included in guide	eBay	See guide
Brake Levers / Sensors	1 set	Cut-off sensor type in guide	eBay / OEM	See guide
Fuse / Safety Hardware	Per spec	Full list and ratings in guide	Hardware	See guide
WHEELS & BRAKES				
Wheel Set (Front + Rear)	1 set	Gold supermoto wheels — new. Sizing and source spec in guide	New	See guide
Brake System	1 set	OEM retained or upgraded — spec in guide	eBay / OEM	See guide
Tires	F+R	Per your use case — recommendations in guide	eBay / local	See guide

TOTAL ESTIMATED BUILD COST

Parts Pricing Disclaimer: All costs are estimated based on eBay and specialty supplier pricing at time of original build. Market prices fluctuate. The full guide includes exact part numbers, search terms, and sourcing strategies to hit the under-\$9K target. Prices may vary.



52KW CRF520E — FULL BUILD OVERVIEW

CRF520E ELECTRIC SUPERMOTO BUILD

520E

52KW | JACKSHAFT DRIVE | 17" SUPERMOTO

BUILD HIGHLIGHTS

- Ventus 72V 52KW Motor (E-520)
- Dual Stage Reduction (4:1 / 3.85:1)
- Custom Jackshaft Assembly
- 3.5" Extended Swingarm
- Custom Subframe + Seat
- Full Sur-Ron Harness
- 72V Battery System
- Baja Designs Lighting
- Full Handguards
- Supermoto 17" Wheels
- Hologram Graphics Kit

SPECIFICATIONS

Motor: Ventus 72V 52KW (E-520)

Controller: Ventus 72V

Battery: 72V High Performance

Primary Reduction: 12T / 48T (4:1)

Secondary Reduction: 14T / 54T (3.85:1)

Top Speed (Est): 85+ MPH

Rear Wheel: 17" Supermoto

Front Wheel: 17" Supermoto

Brakes: Front & Rear Hydraulic

Weight (Est): ~220 lbs

PARTS OVERVIEW

- CRF520E Frame
- 3.5" Extended Swingarm
- Lowering Link
- Custom Subframe
- Triple Clamp & Bars
- Ventus 72V Motor Kit
- Jackshaft Assembly
- 420 Chain Conversion
- KKE 17" Supermoto Wheels
- Sur-Ron Harness
- Baja Designs Lighting
- OEM Grips / Footpegs
- Full Plastics + Hologram Graphics

52KW

PEAK POWER

DUAL STAGE REDUCTION

4:1 / 3.85:1

JACKSHAFT DRIVE

420 CHAIN CONVERSION

17" SUPERMOTO

17" SUPERMOTO WHEELS

BAJA DESIGNS LIGHTING

52 PRO HEADLIGHT

BUILT TO PERFORM

72V BATTERY SYSTEM

FRONT VIEW

REAR VIEW

JACKSHAFT SYSTEM (INSIDE VIEW)

PRIMARY

12T / 48T (4:1)

SECONDARY

14T / 54T (3.85:1)

DETAILS

BAJA DESIGNS 52 PRO HEADLIGHT

FULL HANDGUARDS RENTAL

VENTUS 72V 52KW MOTOR

JACKSHAFT INSIDE VIEW

3.5" EXTENDED SWINGARM

TOP VIEW

ANGLE VIEW

BUILD TOTAL (ESTIMATED)

\$8,659+ USD

VENTUS ELECTRIC

KKE WHEELS

BAJA DESIGNS

RENTAL

SPARTAN RACING AHEAD

DC

WARP9 RACING

SCAR

CRF250R 21KW — ALTERNATIVE BUILD

CRF250R ELECTRIC SUPERMOTO BUILD

250R

21KW | JACKSHAFT DRIVE | 17" SUPERMOTO

HIGHLIGHTS

- Ventus 72V 21kW Motor
- Dual Stage Reduction (4:1 / 3.85:1)
- Custom Jackshaft Assembly
- 3.5" Extended Swingarm
- Custom Subframe
- Full Sur-Ron Harness
- 72V Battery System
- Baja Designs Lighting
- Supermoto 17" Wheels

SPECIFICATIONS

Motor: Ventus 72V 21kW

Controller: Ventus 72V

Battery: 72V High Performance

Primary Reduction: 12T / 48T (4:1)

Secondary Reduction: 14T / 54T (3.85:1)

Top Speed (Est): 75+ MPH

Rear Wheel: 17" Supermoto

Front Wheel: 17" Supermoto

Brakes: Front & Rear Hydraulic

Weight (Est): ~220 lbs

PARTS OVERVIEW

- CRF250R Frame
- Extended Swingarm
- Lowering Link
- Custom Subframe
- Triple Clamp & Bars
- Ventus 72V Motor Kit
- Jackshaft Assembly
- 420 Chain Conversion
- KKE 17" Supermoto Wheels
- Sur-Ron Harness
- Baja Designs Lighting
- OEM Grips / Footpegs
- Full Plastics + Graphics Kit

21KW

PEAK POWER

DUAL STAGE REDUCTION

4:1 / 3.85:1

JACKSHAFT DRIVE

420 CHAIN CONVERSION

17" SUPERMOTO

17" SUPERMOTO WHEELS

BAJA DESIGNS LIGHTING

52 PRO HEADLIGHT

BUILT TO PERFORM

72V BATTERY SYSTEM

FRONT VIEW

REAR VIEW

JACKSHAFT SYSTEM (TOP VIEW)

PRIMARY

12T / 48T (4:1)

SECONDARY

14T / 54T (3.85:1)

DETAILS

LOWERING LINK

CUSTOM JACKSHAFT

CNC CHAIN ADJUSTER

TOP VIEW

ANGLE VIEW

BUILD TOTAL (ESTIMATED)

\$8,659+ USD

VENTUS ELECTRIC

KKE WHEELS

BAJA DESIGNS

RENTAL

OEM

WARP9 RACING

SCAR

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SHARK NOT GOLDFISH · LEAVE THE SCHOOL
BEHIND

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Electric Build
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E-520 JACKSHAFT — FABRICATION & ASSEMBLY GUIDE

E-520 JACKSHAFT

FABRICATION & ASSEMBLY GUIDE

DUAL-STAGE REDUCTION SYSTEM BUILT STRONG. BUILT TO LAST.

SYSTEM OVERVIEW

PRIMARY
12T → 48T
(4.00:1)

SECONDARY
14T → 54T
(3.86:1)

FROM MOTOR 12T SPROCKET

TO REAR WHEEL 54T SPROCKET

TOTAL REDUCTION: 15.44:1

CRITICAL DIMENSIONS	
SHAFT DIAMETER	20mm
BEARING TYPE	UCP204
BEARING CENTER TO CENTER	197mm (7.76")
USABLE SHAFT LENGTH	240mm (9.45")
PLATE THICKNESS	3/8" (9.5mm)
CHAIN TYPE	#420 O-RING (RK 420SB)
CHAIN LINE TOLERANCE	±1.0mm MAX

TORQUE SPECS	
BOLT SIZE	TORQUE
M8 (SPROCKET)	22 Nm (16 ft-lb)
M10 (BEARING/PLATE)	40-45 Nm (30-33 ft-lb)
SET SCREWS / COLLARS	10-12 Nm (8-9 ft-lb)

USE BLUE THREADLOCKER

PLATE MATERIAL

3/8" (9.5mm)
MILD STEEL (A36/1018)

KEY FEATURES

- ADJUSTABLE SLOTTED MOUNTING HOLES
- PRECISION CHAIN ALIGNMENT
- HIGH STRENGTH & RIGID
- SERVICEABLE DESIGN
- BUILT FOR EXTREME TORQUE

E520

ELECTRIC BUILD

DESIGNED TO WIN. BUILT TO DESTROY.

1 JACKSHAFT ASSEMBLY — EXPLODED VIEW

- M8 SPROCKET BOLTS
- 12T SPROCKET (MOTOR SIDE)
- 5mm SPACER
- SHAFT COLLAR (20mm ID)
- UCP204 BEARING (LEFT)
- 3mm INNER SPACER
- 48T SPROCKET (PRIMARY)
- 20mm KEYSHAFT
- 3mm SPACER
- UCP204 BEARING (RIGHT)
- 5mm SPACER
- 14T SPROCKET (OUTPUT)
- 54T SPROCKET (REAR SIDE)
- SHAFT COLLAR
- RETAINER / LOCK PLATE
- M10 SHAFT BOLT

2 STEP-BY-STEP ASSEMBLY

- BUILD SHAFT OFF BIKE**
Assemble all components on the shaft in order. Leave all bolts and collars loose.
- MOUNT BEARINGS TO PLATE**
Place bearings in slotted holes. Install M10 bolts loosely to allow movement.
- INSTALL ON FRAME (LOOSE)**
Place plate on frame. Do NOT tighten. Use clamps to hold in place.
- ALIGN SECONDARY CHAIN**
Align 14T sprocket to rear sprocket perfectly. Adjust bearing position in slots.
- ALIGN PRIMARY CHAIN**
Align 48T sprocket to motor sprocket. Use spacers as needed for perfect chain line.
- FINAL TIGHTEN & MARK**
Tighten all bolts to spec. Mark frame through mounting holes. Remove and drill.

IMPORTANT NOTES

- ALWAYS ALIGN SECONDARY (REAR) CHAIN FIRST.
- DO NOT FULLY TIGHTEN UNTIL BOTH CHAINS ARE PERFECT.
- CHECK CHAIN ALIGNMENT AFTER EVERY ADJUSTMENT.
- USE STRAIGHT EDGE OR LASER FOR ACCURATE ALIGNMENT.

CHAIN LINE TOLERANCE

±1.0mm MAX



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GET THE FULL GUIDE

This document is the **product overview and parts index**. The complete Guide #003 includes all detailed assembly steps, exact part numbers, wiring diagrams, tuning baselines, fitment notes, and builder's commentary.

\$249

GUIDE #003 — 52KW
CRF450 BUILD

\$149

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PROTOTYPE

\$399

ALL 4 GUIDES —
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HOW TO ORDER

Email **sharknotgoldfish@gmail.com** with the guide number(s) you want. We'll send a payment link (Venmo / PayPal / Zelle) and the PDF guide within 24 hours of payment confirmation. Questions? Same email.

ABOUT SNG CYCLES

SNG Cycles is a custom electric motorcycle brand founded by Bryan Jaworski in Royal Palm Beach, Florida. Every build in the SNG lineup — and every guide in the Builder Series — came out of a garage, not a boardroom. Fourteen months. Three R&D prototypes. Over \$50K of personal investment. These guides exist because the information needed to build properly didn't exist anywhere else, so the builder documented everything.

Shark Not Goldfish. Leave the School Behind.

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